

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
sns.set_theme(style='whitegrid')
```

```
data = {
    "Athlete": ["Usain Bolt", "Michael Phelps", "Simone Biles", "Neeraj Chopra",
               "Carl Lewis", "Katie Ledecky", "Abhinav Bindra", "Mo Farah"],
    "Country": ["Jamaica", "USA", "USA", "India", "USA",
               "USA", "USA", "India", "UK", "Jamaica"],
    "Sport": ["Athletics", "Swimming", "Gymnastics", "Athletics", "Athletics",
             "Swimming", "Swimming", "Shooting", "Athletics", "Athletics"],
    "Gold": [8, 23, 7, 1, 7, 9, 7, 1, 4, 3],
    "Silver": [0, 3, 2, 0, 3, 1, 3, 0, 0, 4],
    "Bronze": [0, 2, 2, 0, None, 0, 1, 0, 0, 0]
}
```

```
df = pd.DataFrame(data)
```

```
print(df.shape)
print(df.dtypes)
print(df.head(3))
```

```
(10, 6)
Athlete      object
Country      object
Sport        object
Gold         int64
Silver       int64
Bronze       float64
dtype: object
```

	Athlete	Country	Sport	Gold	Silver	Bronze
0	Usain Bolt	Jamaica	Athletics	8	0	0.0
1	Michael Phelps	USA	Swimming	23	3	2.0
2	Simone Biles	USA	Gymnastics	7	2	2.0

```
print(df.isnull().sum())
```

```
Athlete      0
Country      0
Sport        0
Gold         0
Silver       0
Bronze       1
dtype: int64
```

```
df['Bronze'] = df['Bronze'].fillna(0)
```

```
df['Bronze'] = df['Bronze'].astype('int')
```

```
print('Missing after cleaning:', df.isnull().sum().sum())
print(df.describe())
```

```
Missing after cleaning: 0
```

	Gold	Silver	Bronze
count	10.000000	10.000000	10.000000
mean	7.000000	1.600000	0.500000
std	6.306963	1.577621	0.849837
min	1.000000	0.000000	0.000000
25%	3.250000	0.000000	0.000000
50%	7.000000	1.500000	0.000000
75%	7.750000	3.000000	0.750000
max	23.000000	4.000000	2.000000

```
df['Total'] = df[['Gold', 'Silver', 'Bronze']].sum(axis=1)
```

```
top_athlete = df.sort_values(by='Total', ascending=False).iloc[0]
print("\nTop Athlete:\n", top_athlete)
```

```
Top Athlete:
Athlete    Michael Phelps
Country           USA
Sport        Swimming
Gold           23
Silver          3
Bronze          2
Total          28
Name: 1, dtype: object
```

```
country_total = df.groupby('Country', observed=False)['Total'].sum()
print("\nTotal Medals by Country:\n", country_total)
```

```
Total Medals by Country:
Country
India      2
Jamaica    15
UK          4
USA        70
Name: Total, dtype: int64
```

```
sport_count = df.groupby('Sport', observed=False)['Sport'].count()
print("\nSport Counts:\n", sport_count)
```

```
Sport Counts:
Sport
```

```
Athletics      6
Gymnastics     1
Shooting       1
Swimming       2
Name: Sport, dtype: int64
```

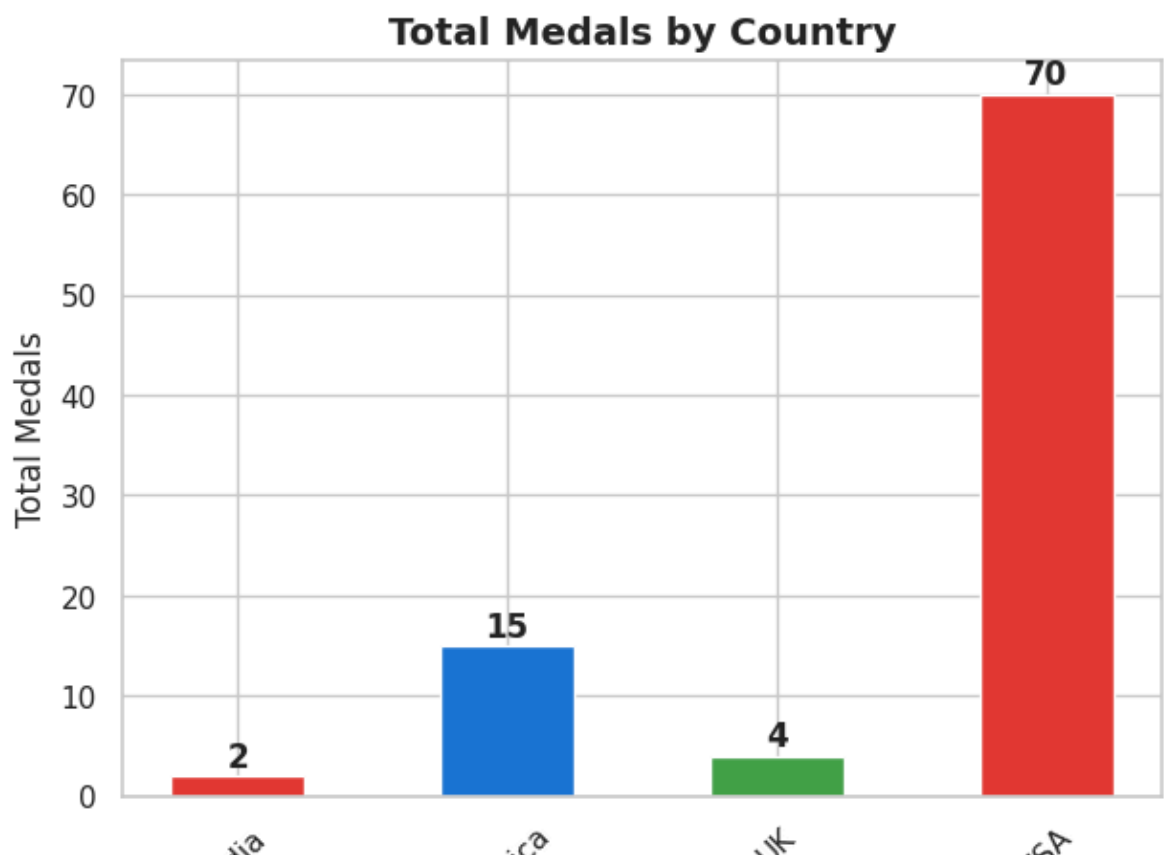
```
avg_gold = df['Gold'].mean()
print(f"\nAverage Gold Medals: {avg_gold:.2f}")
```

Average Gold Medals: 7.00

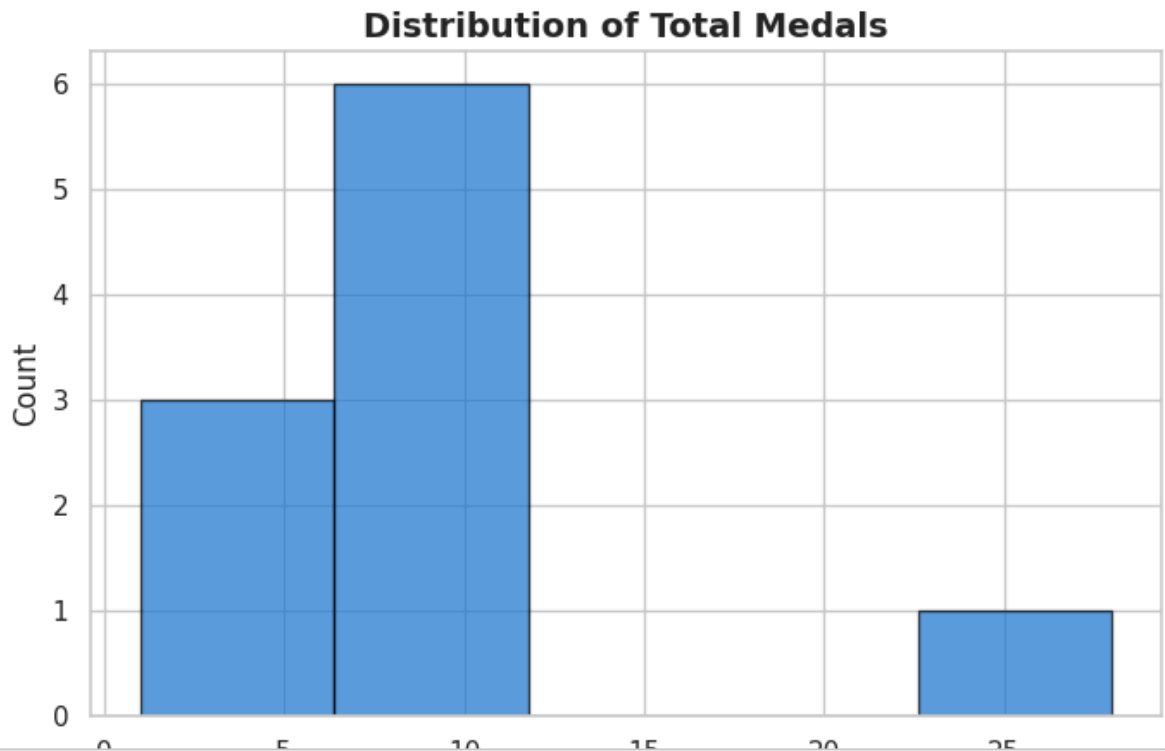
```
india_total = df[df['Country'] == 'India']['Total'].sum()
print("\nIndia Total Medals:", india_total)
```

India Total Medals: 2

```
fig, ax = plt.subplots(figsize=(7, 5))
bars = ax.bar(country_total.index, country_total.values,
               color=['#E53935', '#1976D2', '#43A047'], width=0.5, edgecolor='black')
for bar, val in zip(bars, country_total.values):
    ax.text(bar.get_x()+bar.get_width()/2, val+1,
            f'{val}', ha='center', fontweight='bold')
ax.set_title('Total Medals by Country', fontsize=14, fontweight='bold')
ax.set_ylabel('Total Medals')
plt.xticks(rotation=45)
plt.show()
```



```
fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(df['Total'], bins=5, alpha=0.7, color='#1976D2', edgecolor='b')
ax.set_title('Distribution of Total Medals', fontsize=14, fontweight='bold')
ax.set_xlabel('Total Medals')
ax.set_ylabel('Count')
plt.show()
```



```
df_sorted = df.sort_values(by='Total', ascending=False)
fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(df_sorted['Athlete'], df_sorted['Total'], marker='o', color='b')
for x, y in zip(df_sorted['Athlete'], df_sorted['Total']):
    ax.text(x, y+0.5, str(y), ha='center')
ax.set_title('Total Medals by Athlete (Line Chart)', fontsize=14, fontweight='bold')
ax.set_xlabel('Athlete')
ax.set_ylabel('Total Medals')
plt.xticks(rotation=45)
plt.grid()
plt.show()
```

